

Venkata Veekshith

23ee01008@iitbbs.ac.in | github.com/Veekshith2070 | linkedin.com/in/bondada-venkata-veekshith | Portfolio

Profile

B.Tech Electrical Engineering student at IIT Bhubaneswar with a strong foundation in **Python, data engineering, algorithmic problem solving and AI workflows**. Experienced in building time-series processing pipelines, **PyTorch** deep learning models, and designing scalable backend architectures with relational databases and REST APIs.

Education

Indian Institute of Technology Bhubaneswar

B.Tech in Electrical Engineering

Sasi Educational Institute

Senior School Certificate Examination, CBSE

CGPA: 8.54/10

Aug 2023 – Present

94.4%

Apr 2021 – Mar 2023

Experience

Research Intern – Brain Signal Processing / EEG-iEEG Research Lab, IIT Bhubaneswar *Jun 2026 – Jul 2026*

- Engineered synthetic paired **20-channel EEG and 12-channel iEEG datasets for 18 subjects** using latent neural source modeling and automated signal generation pipelines.
- Developed IED and Non-IED event generation pipelines with source-level signal injection and automated event-window extraction using Python and NumPy.
- Evaluated signal realism across subjects using **correlation analysis, power spectral density (PSD) and inter-subject spectral similarity metrics**.
- Implemented and evaluated a **VAE-cGAN** pipeline incorporating SPADE residual blocks, LSTM layers and PatchGAN for cross-modality EEG-to-iEEG signal reconstruction.

Projects

IoT Sensor Telemetry & Anomaly Detection Pipeline

Python, Pandas, NumPy, Time-Series Analytics

GitHub

- Built an end-to-end data processing pipeline to ingest, clean, and analyze high-frequency multi-sensor IoT telemetry (temperature, vibration, pressure).
- Implemented missing value imputation strategies and rolling-window statistical aggregations to track device degradation patterns.
- Designed statistical Z-score thresholding algorithms to detect and log operational anomalies and thermal spikes in real time.
- Structured cleaned telemetry data into feature matrices for downstream diagnostic querying and retrieval workflows.

SplitWise Clone – Expense Management Application

Node.js, Express.js, PostgreSQL, REST APIs, React.js

GitHub | Live Demo

- Designed a relational **PostgreSQL schema** utilizing primary/foreign keys and junction tables for users, groups, expenses and settlements.
- Implemented balance calculations and a robust **settlement state machine** supporting pending, accepted, and rejected settlement requests.
- Engineered secure RESTful APIs with **JWT authentication**, protected endpoints, and authorization middleware; deployed on **Render**.
- Designed group and expense management workflows supporting member validation, equal/custom expense splitting, and exact paise-level arithmetic.

Technical Skills

Languages: Python, C++, SQL, JavaScript

Data Science & AI: Pandas, NumPy, PyTorch, SciPy, RAG & Retrieval (LangChain, ChromaDB), MNE-Python, MATLAB

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, OS Concepts

Backend & Databases: REST APIs, Node.js, Express.js, PostgreSQL, SQLite, JWT

Tools & Platforms: Git, GitHub, Linux/Bash, VS Code, Google Colab, Render

Frontend: HTML, CSS, React.js

Certifications

NxtWave: Python | JavaScript | Node.js & Express.js | Responsive Web Development

Positions of Responsibility

Core Team Member – Pravaah, IIT Bhubaneswar

Sep 2024 – May 2025

- Delegated tasks to junior members and volunteers while coordinating event responsibilities.
- Negotiated sponsorship deliverables, sponsorship tiers and monetary/in-kind contributions.

Associate – E-Summit, IIT Bhubaneswar

Oct 2023 – Sep 2024

- Collaborated with the core team to execute sponsor deliverables including branding, social media mentions and on-site visibility.